



Accenture Network Security — Tier 2 Handbook & MCQ Bank

Comprehensive Technical Study Notes + 25 Verified Exam MCQs with In-Depth Solutions

Tier 2 Priority: Next-highest priority after cryptography, attacks, and security-device topics. Recommended study sequence:

1. Authentication vs Authorization
2. Access Control Models (DAC, MAC, RBAC, ABAC)
3. VPN & Encrypted Tunneling
4. SSL / TLS Protocols
5. HTTPS & Port Mappings
6. CIA Triad
7. Access Control Lists (ACL)
8. E-mail Bombing
9. Buffer Overflow

1. Authentication

Definition: The process of verifying the identity of a user, device, or system. In simple words: *Authentication* = "Who are you?"

Username + Password → System checks credentials → "You are Abhishek" (Authentication Successful)

Three Common Authentication Factors:

- **1. Something you know (Knowledge Factor):** Password, PIN, Security Question, Passphrase.
- **2. Something you have (Ownership Factor):** OTP (SMS/App), Smart card, Mobile phone, Security hardware token.
- **3. Something you are (Inherence / Biometric Factor):** Fingerprint, Face recognition, Iris scan, Voice biometric.

2. Multi-Factor Authentication (MFA)

MFA requires **two or more different authentication factor categories**.

Valid MFA Examples:

- Password (*Know*) + OTP (*Have*) → **MFA** ✓
- Password (*Know*) + Fingerprint (*Are*) → **MFA** ✓
- Smart Card (*Have*) + Iris Scan (*Are*) → **MFA** ✓

Important Distinction: Two passwords or a Password + PIN are *NOT* MFA because both belong to "Something you know".

3. Authentication vs Authorization ★★★★★

This is one of the highest-yield distinctions in technical assessments.

- **Authentication:** Verifies identity ("Who are you?"). Happens first.
- **Authorization:** Determines permissions and access rights ("What can you access/do?"). Happens after authentication.

Feature	Authentication	Authorization
Core Question	"Who are you?"	"What can you do / access?"
Purpose	Verifies Identity	Verifies Permissions / Privileges
Order of Execution	Happens first (Step 1)	Comes after identity verification (Step 2)
Primary Mechanism	Username, Passwords, Biometrics, OTPs	Roles, Permissions, Policies, ACLs, Tokens
Failure Scenario	"Invalid username or password"	"Access Denied: You lack permissions"

Key Scenario: A user enters the correct username and password but receives "Access Denied: You don't have permission to delete users." → **Authentication succeeded; Authorization failed.**

4. Access Control Models

Access control decides who can access what resources based on defined policies.

- **DAC (Discretionary Access Control):** The **owner** of the resource decides who can access it. (Key word: *Owner decides*).
- **MAC (Mandatory Access Control):** Access is controlled by a **central security policy** based on security classifications (Top Secret, Secret, Confidential, Public). Users cannot change permissions. (Key word: *System/Security policy decides*).
- **RBAC (Role-Based Access Control) ★:** Permissions are assigned based on **job roles** (e.g., Student vs Professor vs Admin). (Key word: *Role decides access*).
- **ABAC (Attribute-Based Access Control):** Access depends on multiple dynamic **attributes**: User + Role + Location + Device + Time + Resource. (Key word: *Contextual attributes decide*).

Model	Main Basis	Accenture Exam Trap / Identifier
DAC	Owner Discretion	"A user creates a file and shares it with a coworker."
MAC	Security Policy / Clearance	"Military classification: Top Secret vs Confidential."
RBAC	Job Role	"Permissions are assigned based on job role (Manager, Employee)."
ABAC	Dynamic Attributes	"Access restricted to company laptops during working hours."

5. Virtual Private Network (VPN) ★★★★★

VPN creates a secure, encrypted connection (tunnel) over a public untrusted network (the Internet).

Remote Employee —[Encrypted VPN Tunnel]—> Internet —> Secure Corporate Server

- **Primary Purposes:** Secure remote corporate access, protect data in transit across untrusted Wi-Fi, connect branch offices.
- **Exam Note:** VPN does *not* make you completely anonymous; it primarily provides secure communication over untrusted networks.

6. SSL vs TLS & HTTPS ★★★★★

- **SSL (Secure Sockets Layer):** Older, deprecated protocol with known cryptographic vulnerabilities.
- **TLS (Transport Layer Security):** Modern cryptographic standard that replaced SSL (TLS 1.2, TLS 1.3).
- **What TLS Provides:**
 1. **Confidentiality:** Strong encryption prevents eavesdropping.
 2. **Integrity:** Cryptographic hashes prevent data modification.
 3. **Authentication:** Digital certificates verify server identity.
- **HTTPS (HTTP over TLS):** Encrypts HTTP traffic. *Formula: HTTP + TLS = HTTPS*. Operates on default **Port 443** (Standard HTTP uses **Port 80**).
- **Security Trap:** HTTPS encrypts transit communication, but does *not* guarantee the website itself is safe (e.g., a phishing site can have an HTTPS certificate).

Protocol	Default Port	Transport Security	Data Integrity & Auth
HTTP	80	✗ Cleartext (Unencrypted)	✗ No integrity or authentication
HTTPS	443	✓ TLS Encryption	✓ Cryptographic certificates & hashing
FTP	21	✗ Cleartext File Transfer	✗ Unencrypted
SMTP	25	✗ Cleartext Mail Routing	✗ Needs STARTTLS upgrade
SSH	22	✓ Secure Shell Encryption	✓ Public key / Password Auth
DNS	53	✗ Standard UDP/TCP	✗ Plain text query responses

7. The CIA Triad ★★★★★

- **C — Confidentiality:** Only authorized individuals can view information (Protected via Encryption, MFA, ACLs). *"Can you see it?"*
- **I — Integrity:** Data remains accurate and unaltered without authorization (Protected via Hashing, Digital Signatures, MACs). *"Is it unchanged?"*
- **A — Availability:** Systems and resources are accessible to authorized users when needed (Protected via Redundancy, Load Balancing, DDoS Protection). *"Is it accessible?"*

Fast Scenario Identifier:

- Attacker steals/reads confidential emails → **Confidentiality Violated**
- Attacker alters bank transfer from ₹500 to ₹50,000 → **Integrity Violated**
- Attacker takes web server down with DoS/DDoS → **Availability Violated**

8. Access Control Lists (ACL) ★★★★★

An ACL contains sequential rules (Permit / Deny) evaluated by routers and firewalls to filter traffic.

- **Standard ACL:** Filters traffic primarily based on **Source IP address only** (Evaluated closest to destination).
- **Extended ACL:** Filters traffic based on **Source IP, Destination IP, Protocol (TCP/UDP/ICMP), and Port number** (Evaluated closest to source).

9. E-mail Bombing vs Buffer Overflow

- **E-mail Bombing:** A resource-exhaustion attack flooding an inbox with thousands of emails to overwhelm storage, crash servers, or obscure fraud alerts. (*Bombing = Flooding volume; Phishing = Tricking credentials*).
- **Buffer Overflow:** Occurs when a program writes more data into a fixed-size memory buffer than allocated, overwriting adjacent memory. Can cause application crashes, memory corruption, and arbitrary code execution.

```
Allocated: char buffer[10];  ← Input given: 100 bytes (Overwrites stack/registers)
```

🔥 Accenture Tier-2 MCQ Practice Bank (Q1 – Q25)

Each question includes the direct answer, full conceptual rationale, breakdown of incorrect options, and memory tricks.

Q1 Authentication

A user enters a username and password into an application. The system checks whether the credentials belong to that user. What security process is being performed?

- A. Authorization | B. Authentication | C. Encryption | D. Access control

✓ Answer: B. Authentication

Explanation: Verifying credentials against recorded user identities answers the fundamental question: *"Who are you?"*

✗ A (Authorization): Determines permissions after identity is confirmed. ✗ C (Encryption): Data obfuscation. ✗ D (Access control): Broader policy enforcement.

Q2 Authentication vs Authorization

A user successfully logs into an organization's system but receives an error when trying to delete employee records because their role does not have permission. What failed?

- A. Authentication | B. Encryption | C. Authorization | D. Identification

✓ Answer: C. Authorization

Explanation: Authentication succeeded because login worked. The subsequent check for delete permissions failed at the authorization layer.

🧠 **Memory Rule:** Authentication = Identity proof | Authorization = Permission check.

Q3 Multi-Factor Authentication (MFA)

A banking application requires a customer to enter a password and then enter an OTP sent to their registered mobile phone. This is an example of:

- A. Single-factor authentication | B. Multi-factor authentication | C. Authorization | D. Encryption

✓ Answer: B. Multi-factor authentication

Explanation: Password (Something you know) + OTP on phone (Something you have) spans two distinct factor categories, creating valid 2FA/MFA.

Q4 Authentication Factor Types

Which of the following is an example of "something you are"?

- A. Password | B. PIN | C. Fingerprint | D. Security token

✓ Answer: C. Fingerprint

Explanation: Biometric traits (fingerprint, retina, face) belong to "Something you are". Passwords/PINs are "Something you know"; tokens are "Something you have".

Q5 Role-Based Access Control (RBAC)

A company gives employees permissions according to their job roles (Manager → View+Modify, Employee → View, Admin → Manage). Which access control model is being used?

- A. DAC | B. MAC | C. RBAC | D. ABAC

✓ Answer: C. RBAC

Explanation: Role-Based Access Control binds system privileges directly to organizational role definitions.

Q6 Discretionary Access Control (DAC)

A user creates a document and decides which other users can read or modify it. Which access control model does this best represent?

- A. MAC | B. DAC | C. RBAC | D. ABAC

✓ **Answer: B. DAC**

Explanation: Under Discretionary Access Control, the object creator/owner possesses full discretion to configure permissions for others.

Q7 Mandatory Access Control (MAC)

A military organization classifies documents as Top Secret, Secret, Confidential, and Public. Access is controlled according to centrally defined security classifications. Which model is this?

- A. DAC | B. RBAC | C. MAC | D. ABAC

✓ **Answer: C. MAC**

Explanation: MAC enforces strict, system-mandated labels and clearances that cannot be overridden by individual resource owners.

Q8 Attribute-Based Access Control (ABAC)

A company's system allows access to a sensitive application only when: The user is an employee, device is company-approved, user is inside corporate network, and request occurs during working hours. Which model best fits?

- A. DAC | B. MAC | C. RBAC | D. ABAC

✓ **Answer: D. ABAC**

Explanation: Dynamic multi-variable policy evaluation (User + Device + Network Location + Time) is the exact definition of Attribute-Based Access Control.

Q9 VPN Applications

An employee working from home needs secure access to the company's internal network over the public Internet. Which technology is most appropriate?

- A. VPN | B. DNS | C. FTP | D. DHCP

✓ **Answer: A. VPN**

Explanation: Virtual Private Networks establish an encrypted tunnel over public infrastructure directly into the enterprise intranet.

Q10 Primary Purpose of VPN

What is the primary purpose of a VPN in a remote-access scenario?

- A. Increase Internet speed | B. Create a secure connection over an untrusted network | C. Assign IP addresses | D. Replace the firewall

✓ **Answer: B. Create a secure connection over an untrusted network**

Explanation: VPNs encapsulate and encrypt packets to provide confidential transit over insecure public networks.

Q11 Web Communication Security

A developer wants to protect HTTP communication between a browser and web server from eavesdropping and tampering. Which should be used?

- A. HTTP | B. HTTPS | C. FTP | D. SMTP

✓ **Answer: B. HTTPS**

Explanation: HTTPS applies Transport Layer Security (TLS) to standard HTTP traffic, securing it against sniffing and interception.

Q12 Standard HTTPS Port

A network administrator wants to allow normal HTTPS traffic through a firewall. Which port should generally be allowed?

A. 21 | B. 25 | C. 80 | D. 443

✓ **Answer: D. 443**

Explanation: Default TCP port for HTTPS is 443 (HTTP = 80, FTP = 21, SMTP = 25).

Q13 SSL vs TLS Evolution

Which statement is correct?

A. TLS is the older protocol replaced by SSL | B. SSL replaced TLS | C. TLS is the modern successor to SSL | D. TLS is an encryption algorithm like AES

✓ **Answer: C. TLS is the modern successor to SSL**

Explanation: Transport Layer Security (TLS) is the modernized protocol designed to remediate deprecated SSL protocols.

Q14 HTTPS Composition

Which statement best describes HTTPS?

A. HTTPS is HTTP combined with TLS protection | B. HTTPS is an encryption algorithm | C. HTTPS is a firewall | D. HTTPS is a hashing algorithm

✓ **Answer: A. HTTPS is HTTP combined with TLS protection**

Explanation: HTTPS runs standard application-layer HTTP messages over a secure cryptographic TLS transport channel.

Q15 CIA: Confidentiality

An attacker intercepts a customer's encrypted communication and somehow obtains the customer's private information. Which CIA property is primarily concerned with preventing unauthorized disclosure?

A. Integrity | B. Availability | C. Confidentiality | D. Authorization

✓ **Answer: C. Confidentiality**

Explanation: Confidentiality guarantees that sensitive information remains concealed from unauthorized observers.

Q16 CIA: Integrity

An attacker modifies a bank transaction from ₹500 to ₹50,000. Which security property has been violated?

A. Confidentiality | B. Integrity | C. Availability | D. Authentication

✓ **Answer: B. Integrity**

Explanation: Unauthorized alteration or data corruption directly breaches the Integrity pillar of information security.

Q17 CIA: Availability

An attacker floods an online shopping website with massive amounts of traffic, causing legitimate customers to lose access. Which CIA property is primarily affected?

A. Confidentiality | B. Integrity | C. Availability | D. Authentication

✓ **Answer: C. Availability**

Explanation: DoS/DDoS attacks exhaust server bandwidth or system resources, denying service availability to legitimate users.

Q18 Encryption & CIA

A company encrypts confidential employee records so that unauthorized people cannot read them. Which CIA property is primarily being protected?

- A. Confidentiality | B. Integrity | C. Availability | D. Authorization

✓ **Answer: A. Confidentiality**

Explanation: Cryptographic encryption ensures data secrecy and protects confidentiality against unauthorized access.

Q19 Access Control Lists (ACL)

A network administrator creates rules that determine whether packets from particular IP addresses are allowed or denied. What is being used?

- A. ACL | B. AES | C. VPN | D. DNS

✓ **Answer: A. ACL**

Explanation: An Access Control List (ACL) comprises ordered permit/deny packet filtering statements on network devices.

Q20 Standard ACL

A router administrator wants to filter traffic primarily based on the source IP address. Which ACL type is generally appropriate?

- A. Standard ACL | B. Extended ACL | C. Dynamic NAT | D. MAC

✓ **Answer: A. Standard ACL**

Explanation: Standard ACLs filter traffic based almost exclusively on Source IP address.

Q21 Extended ACL

A network administrator wants to allow only HTTPS traffic from a specific source network to a specific destination server. Which ACL provides the more appropriate level of filtering?

- A. Standard ACL | B. Extended ACL | C. DAC | D. RBAC

✓ **Answer: B. Extended ACL**

Explanation: Extended ACLs support fine-grained matching across Source IP, Destination IP, Protocol (TCP), and Port (443).

Q22 E-mail Bombing

An attacker sends thousands of emails to a victim's mailbox within a short period, overwhelming the mailbox and making legitimate messages difficult to access. What type of attack is this?

- A. Phishing | B. Email bombing | C. Spoofing | D. Buffer overflow

✓ **Answer: B. Email bombing**

Explanation: Flooding an inbox with massive email volumes to consume mailbox quotas is Email Bombing (Phishing uses deception, Spoofing fakes identity).

Q23 Buffer Overflow Vulnerability

A program has allocated a 10-byte memory buffer but accepts 100 bytes of input without properly checking its size. What vulnerability may occur?

- A. Phishing | B. Buffer overflow | C. DDoS | D. Spoofing

✓ **Answer: B. Buffer overflow**

Explanation: Writing data beyond allocated memory bounds triggers a Buffer Overflow, risking memory corruption.

Q24 Buffer Overflow Exploitation

An attacker provides specially crafted input to a vulnerable application. The input overwrites memory beyond the allocated buffer and potentially changes the program's execution. What type of vulnerability is being exploited?

- A. Buffer overflow | B. SQL injection | C. Phishing | D. Email bombing

✓ **Answer: A. Buffer overflow**

Explanation: Buffer overflow exploits overwrite stack frames or return instruction pointers to divert execution flow.

Q25 Comprehensive Security Architecture Scenario

An organization wants to implement: (1) Prove identity before login, (2) Permissions depend on job role, (3) Remote employees connect through secure tunnel, (4) Web communication uses TLS, (5) Network traffic filtered using permit/deny rules. Which combination is correct?

- A. Authentication + RBAC + VPN + HTTPS/TLS + ACL
B. Authorization + DAC + FTP + HTTP + DNS
C. Authentication + MAC + SMTP + HTTP + DHCP
D. Encryption + RBAC + DNS + FTP + NAT

✓ **Answer: A. Authentication + RBAC + VPN + HTTPS/TLS + ACL**

Component Mapping:

- Prove identity → **Authentication**
- Role-based permissions → **RBAC**
- Remote encrypted tunnel → **VPN**
- Secure web channel → **HTTPS/TLS**
- Permit/Deny traffic rules → **ACL**